Unit Test Report — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
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Prepared by: Komal S Sajja · **Version:** 1.0 · **Date:** 2026-11-02
Build tested: commit `2d74453` · **Baseline:** `Documents/Test_Plan.pdf` v3.1

1. Purpose

This report records the unit-level test cases of the Test Plan (UT_001–UT_068) run against commit `2d74453`. They were run with Django's test runner as part of the full automated suite, and every test name and result below is copied from that run's `test_run.txt`.

2. Environment and procedure

Item	Value
OS	Windows 11 Home (10.0.26300)
Python / Django	Python 3.12.10 / Django 5.2.18
Database	SQLite (Django's temporary in-memory test database)
Build	Commit `2d74453`
Setup	`pip install -r requirements.txt`; `DJANGO_SECRET_KEY=dev`; `python manage.py migrate`
System check	`python manage.py check`: "System check identified no issues (0 silenced)."
Command	`python manage.py test -v 2 2>&1`

3. Summary

The run ended with `Ran 227 tests in 272.197s` and `OK`.

Measure	Count
Tests run (whole suite)	227
Passed / Failed / Errors / Skipped	227 / 0 / 0 / 0
Test Plan UT cases	68
UT cases with an automated test	68
UT-tagged automated tests	69 (UT_068 has two)
Further test reported against UT_060	1 (`test_publishing_before_the_window_closes_is_refused`)

App	Tests
attempts	61
identity	54
examsetup	23
evaluation	22
portal	20
questionbank	15
results	15
repository	10
integrity	7
Total	**227**

4. Results by Test Plan case

The Requirement column is taken from each case's description in Test Plan v3.1. UT_060 and UT_068 have two rows each, one per automated test.

ID	Requirement	Automated test	Result
UT_001	FR-001	test_UT_001_register_with_all_fields_valid (identity.test_auth.RegistrationTests)	Pass
UT_002	FR-001	test_UT_002_roll_number_is_required (identity.test_auth.RegistrationTests)	Pass
UT_003	FR-001	test_UT_003_non_institutional_email_is_rejected (identity.test_auth.RegistrationTests)	Pass
UT_004	FR-001	test_UT_004_invalid_email_is_rejected (identity.test_auth.RegistrationTests)	Pass
UT_005	FR-001	test_UT_005_duplicate_email_is_rejected (identity.test_auth.RegistrationTests)	Pass
UT_006	FR-001	test_UT_006_duplicate_roll_number_is_rejected (identity.test_auth.RegistrationTests)	Pass
UT_007	FR-002	test_UT_007_correct_credentials_establish_a_session (identity.test_auth.SignInTests)	Pass
UT_008	FR-002	test_UT_008_wrong_password_is_refused (identity.test_auth.SignInTests)	Pass
UT_009	FR-002	test_UT_009_unknown_email_has_the_same_message (identity.test_auth.SignInTests)	Pass
UT_010	FR-002	test_UT_010_empty_password_is_refused (identity.test_auth.SignInTests)	Pass
UT_011	FR-003	test_UT_011_every_account_carries_exactly_one_valid_role (attempts.test_cohorts.CohortTests)	Pass
UT_012	FR-004	test_UT_012_admin_creates_a_course (identity.test_courses.CourseTests)	Pass
UT_013	FR-004	test_UT_013_empty_course_title_is_rejected (identity.test_courses.CourseTests)	Pass
UT_014	FR-004	test_UT_014_duplicate_code_and_semester_is_rejected (identity.test_courses.CourseTests)	Pass
UT_015	FR-004	test_UT_015_student_cannot_be_assigned_as_faculty (identity.test_courses.CourseTests)	Pass
UT_016	FR-004	test_UT_016_assigning_same_faculty_twice_has_one_assignment (identity.test_courses.CourseTests)	Pass
UT_017	FR-005	test_UT_017_faculty_can_enroll_a_student (identity.test_courses.CourseTests)	Pass
UT_018	FR-005	test_UT_018_duplicate_enrollment_is_rejected (identity.test_courses.CourseTests)	Pass
UT_019	FR-005	test_UT_019_faculty_account_cannot_be_enrolled_as_student (identity.test_courses.CourseTests)	Pass
UT_020	FR-005	test_UT_020_unassigned_faculty_cannot_enroll_students (identity.test_courses.CourseTests)	Pass
UT_021	FR-006	test_UT_021_student_sees_only_exams_of_enrolled_courses (attempts.test_cohorts.CohortTests)	Pass
UT_022	FR-007	test_UT_022_valid_single_correct_question_is_saved_and_listed (questionbank.tests.QuestionAuthoringTests)	Pass
UT_023	FR-007	test_UT_023_one_option_is_rejected (questionbank.tests.QuestionAuthoringTests)	Pass
UT_024	FR-007	test_UT_024_seven_options_are_rejected (questionbank.tests.QuestionAuthoringTests)	Pass
UT_025	FR-007	test_UT_025_no_correct_option_is_rejected (questionbank.tests.QuestionAuthoringTests)	Pass
UT_026	FR-007	test_UT_026_two_correct_options_are_rejected (questionbank.tests.QuestionAuthoringTests)	Pass
UT_027	FR-008	test_UT_027_missing_metadata_is_rejected (questionbank.tests.QuestionAuthoringTests)	Pass
UT_028	FR-010	test_UT_028_exam_defaults_to_draft (examsetup.tests.ExamAuthoringTests)	Pass
UT_029	FR-010	test_UT_029_duration_must_be_positive (examsetup.tests.ExamAuthoringTests)	Pass
UT_030	FR-010	test_UT_030_exam_policies_are_saved (examsetup.tests.ExamAuthoringTests)	Pass

ID	Requirement	Automated test	Result
UT_031	FR-010	test_UT_031_pass_mark_must_be_between_zero_and_100 (examsetup.tests.ExamAuthoringTests)	Pass
UT_032	FR-012	test_UT_032_draft_exam_is_invisible_to_students (examsetup.test_publication.PublicationTests)	Pass
UT_033	FR-012	test_UT_033_publishing_moves_the_exam_out_of_draft_with_a_server_timestamp (examsetup.test_publication.PublicationTests)	Pass
UT_034	FR-013	test_UT_034_published_exam_window_is_saved_and_shown (examsetup.test_schedule.ExamSchedulingTests)	Pass
UT_035	FR-013	test_UT_035_window_end_must_be_later_than_start (examsetup.test_schedule.ExamSchedulingTests)	Pass
UT_036	FR-014	test_UT_036_exam_starting_tomorrow_is_upcoming_without_start_button (identity.test_dashboard.DashboardTests)	Pass
UT_037	FR-014	test_UT_037_open_window_is_live_with_start_button (identity.test_dashboard.DashboardTests)	Pass
UT_038	FR-014	test_UT_038_submitted_exam_is_completed_without_start_button (identity.test_dashboard.DashboardTests)	Pass
UT_039	FR-015	test_UT_039_start_is_refused_before_the_window_opens (attempts.tests.StartAttemptTests)	Pass
UT_040	FR-015	test_UT_040_start_is_refused_after_the_window_closes (attempts.tests.StartAttemptTests)	Pass
UT_041	FR-015	test_UT_041_start_is_refused_when_a_submitted_attempt_exists (attempts.tests.StartAttemptTests)	Pass
UT_042	FR-017	test_UT_042_remaining_time_uses_server_time_for_helper_and_api (attempts.test_timer_payload.TimerAndPayloadTests)	Pass
UT_043	FR-018	test_UT_043_palette_moves_through_every_state (attempts.test_palette.PaletteTests)	Pass
UT_044	FR-018	test_UT_044_exactly_one_stem_is_rendered (attempts.test_palette.PaletteTests)	Pass
UT_045	FR-019	test_UT_045_response_is_persisted_with_no_save_button (attempts.test_autosave.AutosaveTests)	Pass
UT_046	FR-019	test_UT_046_changing_a_response_keeps_one_row_with_the_new_value (attempts.test_autosave.AutosaveTests)	Pass
UT_047	FR-019	test_UT_047_clearing_a_response_stores_it_empty (attempts.test_autosave.AutosaveTests)	Pass
UT_048	FR-021	test_UT_048_remaining_time_on_resume_counts_from_the_original_start (attempts.test_resume.ResumeTests)	Pass
UT_049	FR-020	test_UT_049_the_paper_submits_itself_at_zero (attempts.test_auto_submit.AutoSubmitTests)	Pass
UT_050	FR-020	test_UT_050_every_response_saved_before_expiry_is_kept (attempts.test_auto_submit.AutoSubmitTests)	Pass
UT_051	FR-020	test_UT_051_no_response_is_accepted_after_auto_submission (attempts.test_auto_submit.AutoSubmitTests)	Pass
UT_052	FR-020	test_UT_052_submitted_at_is_the_start_plus_the_duration (attempts.test_auto_submit.AutoSubmitTests)	Pass
UT_053	FR-023	test_UT_053_correct_response_earns_full_positive_marks (evaluation.tests.MarkingRuleTests)	Pass
UT_054	FR-023	test_UT_054_incorrect_response_loses_the_configured_negative_marks (evaluation.tests.MarkingRuleTests)	Pass
UT_055	FR-023	test_UT_055_unanswered_question_earns_zero_with_no_deduction (evaluation.tests.MarkingRuleTests)	Pass
UT_056	FR-023	test_UT_056_no_deduction_when_negative_marking_is_disabled (evaluation.tests.MarkingRuleTests)	Pass
UT_057	FR-024	test_UT_057_score_exactly_at_the_pass_mark_is_a_pass (evaluation.tests.OutcomeTests)	Pass
UT_058	FR-024	test_UT_058_score_just_below_the_pass_mark_is_a_fail (evaluation.tests.OutcomeTests)	Pass
UT_059	FR-025	test_UT_059_unpublished_result_is_withheld (results.test_publish.PublishTests)	Pass
UT_060	FR-026	test_UT_060_publishing_makes_the_result_visible (results.test_publish.PublishTests)	Pass
UT_060	FR-026	test_publishing_before_the_window_closes_is_refused (results.test_publish.PublishTests)	Pass
UT_061	FR-026	test_UT_061_unpublishing_withdraws_it_and_keeps_the_score (results.test_publish.PublishTests)	Pass
UT_062	FR-027	test_UT_062_scorecard_shows_total_percentage_outcome_and_	Pass

ID	Requirement	Automated test	Result
		every_question (results.test_scorecard.ScorecardTests)	
UT_063	FR-027	test_UT_063_per_question_marks_sum_to_the_total (results.test_scorecard.ScorecardTests)	Pass
UT_064	FR-028	test_UT_064_one_focus_loss_is_one_event_with_a_server_time (integrity.test_focus.FocusTests)	Pass
UT_065	FR-028	test_UT_065_three_losses_are_three_events (integrity.test_focus.FocusTests)	Pass
UT_066	FR-028	test_UT_066_no_loss_means_no_event (integrity.test_focus.FocusTests)	Pass
UT_067	FR-028	test_UT_067_client_supplied_time_is_ignored (integrity.test_focus.FocusTests)	Pass
UT_068	FR-030	test_UT_068_an_edit_records_old_and_new_values (repository.test_audit.AuditTests)	Pass
UT_068	FR-030	test_UT_068_question_edit_records_old_and_new_stem_and_answer_key (repository.test_audit_coverage.AuditCoverageTests)	Pass

5. Defects found

No defects found.

6. Conclusion

All 68 UT cases of Test Plan v3.1 have at least one automated test, and all 70 tests reported against them passed. The whole suite of 227 tests passed with no failures, errors or skips. At the unit level, build 2d74453 behaves as the Test Plan specifies.